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Parametric design of photovoltaic louver integrated shading devices for west facade windows of office buildings in central China

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ABSTRACT

In order to improve the shading performance and power generation effect of Photovoltaic shading integrated devices (PVSDs) on the west facade of buildings in China, a multi-objective optimization design method is proposed in this paper. Taking a west-facing office building in Zhengzhou, Henan Province, as an example, a parametric model of photovoltaic (PV) louvers for windows on the west facade is established, and parametric design research is carried out on the energy-saving potential of PV louver shading systems. Based on the optimization objectives of power generation of PV louvers, indoor thermal discomfort hours, building cooling load and lighting energy consumption, the performance of the west facade office with PVSD under different design strategies was simulated, and the optimal design strategy was determined according to the preference of decision makers. Compared with the basic type without a non-shading model, the indoor lighting energy consumption of the optimized west facade PVSD is slightly increased. However, overall performance is improved, with a 35% reduction in cooling load, a decrease of 804.59 kWh in total energy consumption, and a shortened summer thermal discomfort time of 404 hours. The power generation of PV louvers can replace 46% of the total energy consumption of the room. The results show that the installation of PVSD on the west facade window is important for alleviating the overheating inside the room and improving the energy-saving and power generation potential of the room.

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KEYWORDS
Photovoltaic shading integrated device (PVSD); Multi-objective optimization; Building west facade; Energy consumption; Indoor thermal

1. Introduction

Nowadays, countries worldwide are still in a tense situation of energy crisis and environmental pollution. According to the latest Statistical Review of World Energy 2023, fossil fuels continue to dominate global energy consumption; meanwhile, renewable power, including solar, is generating at record levels, with its capacity meeting 84 percent of net electricity demand growth in 2022 (BP). Therefore, reducing the consumption of fossil energy and accelerating the utilization of renewable power have gradually become the focus of sustainable development of society.

According to official statistics, the energy consumption of China's construction industry accounts for about 33% of the total energy consumption of the whole society (undefined). Office buildings are one of the building types with the highest level of energy consumption compared to other types of buildings, with large construction volumes, large inventories, and high internal staff utilization rates (X. Gao et al. 2017; Sun, Wilson, and Wu 2018). Window is an integral part of the building envelope, whose heat transfer coefficient is much higher than those of other building envelopes. As a result, the thermal performance of windows is poor compared to the rest of the building envelope. It has been demonstrated that energy consumption through window can be responsible for as much as 60% of the total energy consumption of a building (Goia, Haase, and Perino 2013; Ji et al. 2022; Pan, Wei, and Chu 2018; Y. Wang et al. 2022). However, large areas of glass curtain walls and windows are often used in the construction of office buildings due to the pursuit of sufficient indoor light for a good view. The above situation is one of the factors leading to excessive energy consumption in office buildings. Especially in summer, a large amount of solar radiation enters the room through the window system, which causes the temperature inside the building to rise rapidly. These phenomena can easily cause thermal discomfort to indoor workers, while aggravating the problem of energy consumption in office buildings (S. Liu et al. 2019; Teng 2019). In addition, some scholars have found that in some cases, the intensity of solar radiation and indoor heat gain received by office buildings in China through the windows on the west facade are much greater than those on the south facade, and that the westward solar radiation poses a more serious threat to the energy performance of the west-facing rooms of office buildings in China and to the comfort of indoor personnel (Li et al. 2017; Luo et al. 2017; M.

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Wang et al. 2017). Therefore, it is an indispensable way to rationally utilize solar power generation technology to alleviate the current dilemma faced by the west façade rooms of office buildings.

Building Integrated PV (BIPV) is one of the key ways in which the building sector can reduce carbon emissions and support the development of renewable energy (Q. Gao, Yang, and Wang 2022; Su et al. 2022). Combining photovoltaic power generation technology with traditional shading devices for windows can provide buildings with the dual effect of shading and power generation (Luo et al. 2018; W. Zhang, Lu, and Peng 2017). Therefore, applying photovoltaic louver shading devices to the west-facing windows of office buildings in China is an effective means of solving the problem of high energy consumption in rooms on the west façade of some buildings and maintaining stable and comfortable indoor temperatures (Boulahia, Djiar, and Amado 2021; L. Z. Zhang 2019). As one of the specific forms of PV shading integrated device (PVSD), photovoltaic louver shading devices show better applicability and flexibility when facing complex outdoor environments (Hoseinzadeh et al. 2021; Taveres-Cachat et al. 2019). Mandalaki et al (Y. Gao et al. 2018; G. Liu et al. 2022), Hong et al (McNeil et al. 2013; Tools, n.d.). and Zhao et al (Fumo, Mago, and Luck 2010). have investigated the effects of photovoltaic louvers on building performance by means of measurements or simulations, and have shown that photovoltaic louver shading devices have a positive effect on reducing building energy consumption and improving the comfort of indoor personnel. With the deepening of research, the optimization design that only considers a single parameter or objective can no longer meet the needs of social development. Some scholars have begun to use multi-objective optimization design methods to optimize multiple design parameters of photovoltaic louvers to improve building performance (China Academy of Building Research Ltd 2021; Liang, Shi, and Gao 2021; Xie 2018).

This study differs from previous studies mainly in two aspects. Firstly, it is about the selection of design parameters for photovoltaic blades. In previous studies, the blade width is usually set as a fixed value, and the design variables are mostly the number of blades, the rotation angle of the blades, or a combination of the two for optimization. Unfortunately, few of these studies paid attention to the optimization design of photovoltaic blade width. In terms of setting the design variables of the photovoltaic louver shading device, we are not limited to setting the width of the photovoltaic panel slats to a fixed value, but the louver width and the number of blades for the linkage design, i.e., at the same time as the change in the width of the louver, the number of blades also changes to enhance the flexibility of the regulation of the photovoltaic louver system.

Secondly, in terms of the selection of optimization objectives, previous research on the optimization of photovoltaic louvers has mostly focused on the power generation performance of the photovoltaic system and the energy performance of the building, and there is less research on the optimization of combining with the enhancement of indoor light and heat performance. In this study, in addition to focusing on improving the energy performance of the building, the improvement of thermal comfort of indoor workers and the enhancement of indoor natural illumination as the optimization objectives are simultaneously optimized in order to obtain the optimal design strategy under the comprehensive consideration of multiple factors. Moreover, the research on PVSD in China is mostly focused on the south facade of the building (analysis 2015; Redolfi and Khoshtinat 2009; Tzeng and Huang 2011; Yu et al. 2015), and there are relatively few studies on the optimal design of PVSD on the west façade of the building.

Therefore, we aim to propose a multi-objective optimization design method for photovoltaic louver shading devices based on genetic algorithms to solve the dilemma faced by the west facade rooms of office buildings in China and make up for the shortcomings of current research. The main content is divided into the following two aspects: (1) A multi-objective optimization design method for external shading devices of photovoltaic louvers for building windows has been established. The optimization design method uses the number, rotation angle and width of photovoltaic louver blades as design variables, and takes the thermal discomfort hours of indoor occupants, lighting energy consumption, photovoltaic power generation and cooling load as optimization goals; (2) An optimization design study was conducted on the photovoltaic louver device on the west facade window of an office building in central China to improve the illumination, indoor comfort and energy performance of the west facade room of office buildings in China during summer.

This remainder of this work is organized as follows: An overview of the model information, design workflow and methodology used for the study is presented in Section 2, and the various parameters and tools required for the optimization simulations are described in detail. The results of the optimization simulations performed for this study and the selection of the optimal solution are discussed in Section 3. Finally, the conclusions of this study are summarized in Section 4, and the implications for future work are expressed.

2. Methodology

In this study, the Grasshopper parametric platform based on Rhino 3D modeling software was used to establish and simulate the building foundation model, and its reliability has been proven by many researchers (McNeil et al. 2013). Among them, the Ladybug (Tools, n.d.) plug-in with Radiance as the simulation engine and the Honeybee plug-in with EnergyPlus (Fumo, Mago, and Luck 2010) as the simulation engine were used for environmental simulation analysis, such as solar radiation and building energy.

In this paper, RH&GH was used to establish a parameterized model for the research case, and the Honeybee & Ladybug plug-in was used to simulate and analyze the performance of the case. Then, the GA was applied to the optimal design of the scheme to find the Pareto optimal solution set. Finally, the TOPSIS multi-criteria decision-making method was used to analyze the objective function Assign weights to get the best parameter configuration scheme. The detailed description is listed as follows.

2.1. Description of the case study

Zhengzhou City (113:42E, 34:44N) is located in the central part of China, which belongs to the area with good solar energy resources and abundant and stable solar energy resources (Liang, Shi, and Gao 2021). Simulations using Honeybee revealed that the west facade of the building could receive more solar radiation than the south facade in summer, but at the same time, the west-facing rooms were faced with a serious problem of excessive solar radiation intensity.

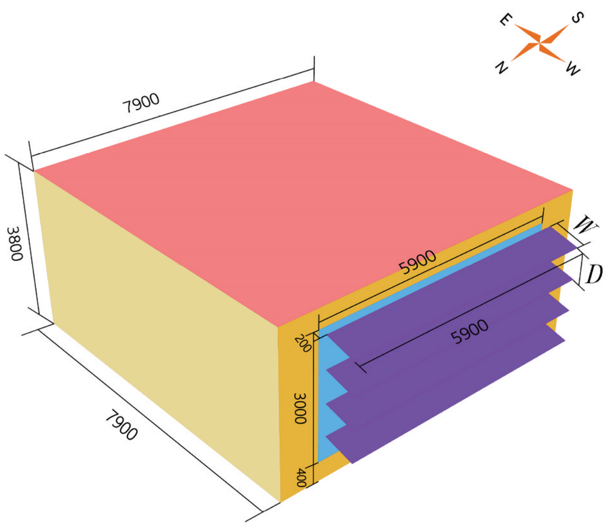


Figure 1. Simulation model.

Table 1. Parameters of building envelope and PV louvers.

Parameters	Value
U-value external wall	0.48
U/SHGC -value external window	1.9/0.47
R-value interior wall	0.32
R-value interior ceiling	0.32
R-value interior floor	0.32
T-value of interior floor	0.76

Table 2. Setting of building operating conditions.

Parameters	Value
Operation time	8: 00–19: 00 on weekdays
Lighting power	13.5
Density of personnel	10
HVAC system	Ideal air load
Set points	26
COP cooling system	1

A west-facing room of an office building in the Zhengzhou area was selected as the base model, and the external sunshade model of PV louvered window openings in the west-facing room was established. In this paper the middle room of the standard floor was selected as the research object to eliminate the influence of the roof, ground, and the building envelope. Figure 1 shows the schematic diagram of the model.

where W is the width of the PV louvers; D is the distance between the PV louvers mm.

The building room size is 7900 mm × 7900 mm × 3800 mm, the window sill height is 400 mm, the window is 3000 mm × 5900 mm, the window-to-wall area ratio is 0.59, and the distance from the PV louvers to the window is 200 mm. The settings of room ventilation, lighting, and personnel cooling are in line with the provisions of the energy-saving design standards for public buildings in cold regions of my country. In order to analyze the impact of west-facing windows on the indoor environment of the building, the ceiling, floor and other three walls were insulated except for the heat transfer on the west-facing exterior wall. The thermal parameters of the building envelope are shown in Table 1. The thermal parameters of the building envelope are shown in Table 1. Artificial lighting is used when the minimum illuminance of the working plane (0.75 m) in the room is 450 lx, and the lighting power is 13.5 W/m^2. The indoor lighting control is continuously adjusted according to the indoor illuminance. In addition, HVAC system is always operational during working hours. The personnel density is 10 m^2/person, the radiation rate is 0.2, and the human body metabolism rate is 1.1 met when the personnel sit and type indoors, and the thermal resistance of the clothing is 0.5 clo in summer working conditions. The outdoor simulated meteorological parameters used are the hour-by-hour meteorological data of the typical meteorological year (CSWD) in Zhengzhou. The building conditions were set according to the Chinese standard GB 55,015–2021 "General Specification for Building Energy Efficiency and Renewable Energy Utilization" (China Academy of Building Research Ltd 2021), as shown in Table 2.

2.2. Description of PV integrated shading devices

The PV material used in this paper is monocrystalline silicon, and the solar energy conversion rate is 20%. The specific parameters are shown in Table 3.

Table 3. Parameters of PV louvers.

Parameters	Value
U-value Solar reflectance of PV louvers (front/rear)	0.234
U-value visible light reflectance of PV louvers (front/rear)	0.133
Solar energy conversion efficiency of PV louvers	20
Overall efficiency of PV system	80

In order to make the PV louvers parametric transformation within a certain range, the total area of the PV louvers was set to be equal to the window area, and the length of the slats was set to be equal to the window width, both of which are 5900 mm. The deflection angle of the PV louvers is between 0° and 90°. The width *W* of the slats of the PV louvers was set to equal the distance *D* of the louvers to ensure that when the PV louvers were deflected by 90°, the PV sunshade device could completely cover the building window. The parameters of PV louver design variables are shown in Table 4, and the schematic diagram of model changes is shown in Table 5.

2.3. Description of the optimization objective and optimization process

Due to the high temperature in Zhengzhou in summer (Xie 2018), the summer period from June 1st to September 30th was selected as the simulation period in this paper. In order to clarify the impact of the PV louver system on the westward window of the building on the indoor thermal environment during this period of time, the building's indoor thermal discomfort hours (in h), i.e., the minimum number of hours when the indoor PMV (Predicted Mean Vote) value exceeds −1 to 1 were selected as the evaluation target of building thermal comfort. The goal of choosing the maximum PV power generation (in kWh/m^2) is to improve the utilization rate of PV energy, thereby alleviating the current energy pressure. In terms of building energy, the two smallest energy consumption in terms of cooling load (in kWh/m^2) and lighting energy consumption □in kWh/m^2) were selected as evaluation indicators. This is because with shades on the windows, less solar radiation enters the solar radiation, and there is less need for indoor cooling, but at the same time, there is an increase in the use of artificial electric lighting. The flow of the simulation was constructed based on building energy simulation, daylight simulation, and PV power generation simulation. Figure 2 illustrates the simulation process and parameter settings of the model for this study.

The simulation process is divided into three steps: 1) Establish the geometric model: input building parameters, climate data, occupancy rate, and geometric data of PV louvers. 2) Perform performance simulation: simulate PV louver power generation, building load, and indoor thermal comfort. 3) Perform multi-objective optimization: use Octopus plug-in for multi-objective optimization of the simulation scheme, and then use to optimize processing, and discuss the Pareto solution set is discussed.

Since the Octopus operator performs multi-objective optimization, the objective value function is optimized for the minimum value. However, when the power generation of PV louvers is optimized, the target criterion should be the maximum. Therefore, when optimizing and discussing the objective function, the PV louver power generation simulation data should be set to a negative value (i.e., the smaller the negative value of power generation after simulation, the better the performance of the PV louver power generation target). Then, the subsequent optimization results were discussed and analyzed.

2.4. Description of the GA optimization process

In this paper, the GA optimization module Octopus was used for optimization operation. The operation process is shown in Figure 3. GA is a method to solve specific problems using principles like the processes of inheritance, mutation, selection, and evolution in natural biology (Redolfi and Khoshtinat 2009, Yu et al. 2015).

The program operation steps of GA are listed as follows: firstly, randomly construct the initial optimization population; secondly, evaluate the population operation results according to the evaluation information of the given target, and select offspring with good adaptability and scale from the results to ensure the diversity of the population; then, perform infinite iterative operations according to the above process cycle; finally, determine one or a group of optimal solutions according to the given target. The input parameters of GA include elite proportion, mutation rate, crossover probability, population size, and

Table 4. Description of design parameters of PV louvers.

Variable	Range
Number of louvers	(BP, undefined, Sun, Wilson, and Wu 2018, X. Gao et al. 2017, Y. Wang et al. 2022, Ji et al. 2022, Pan, Wei, and Chu 2018, Goia, Haase, and Perino 2013, Teng 2019, S. Liu et al. 2019, M. Wang et al. 2017, Li et al. 2017, Luo et al. 2017, Q. Gao, Yang, and Wang 2022, Su et al. 2022) (the interval is 1)
U-value visible light reflectance of PV louvers	[0–90](the interval is 5)

Table 5. Schematic diagram of model changes.

Schematic diagram of model changes.

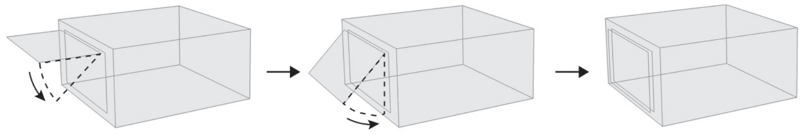
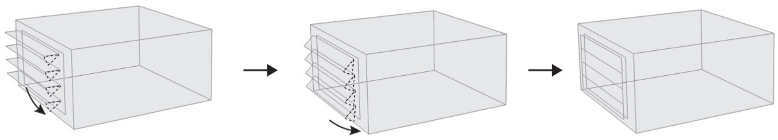
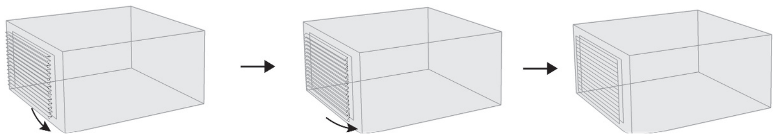
Counts	0°	45°	90°
1			
4			
15			

Figure 2. Flow chart of visual performance optimization simulation in Grasshopper.

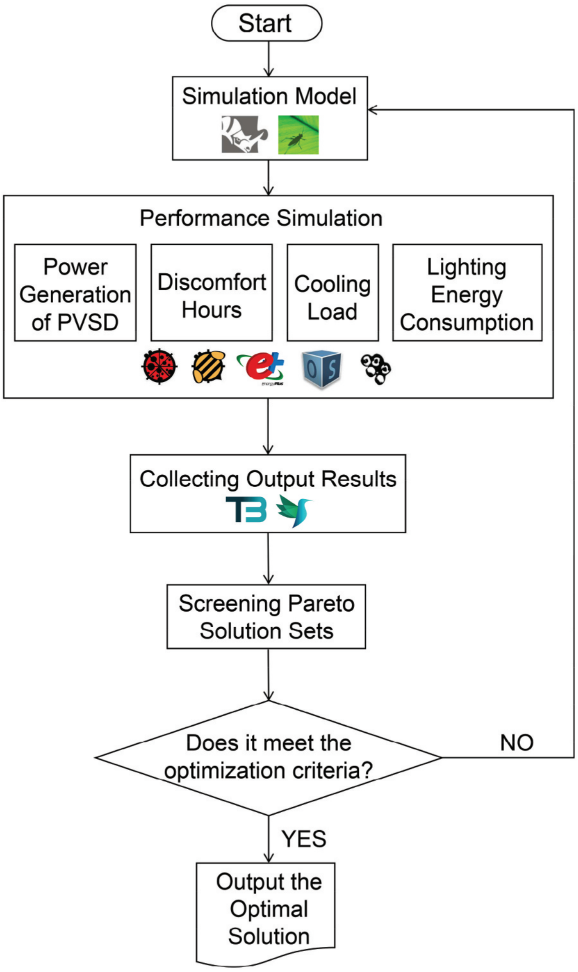


Figure 3. Schematic diagram of the optimization process.

Table 6. GA input parameter settings.

Parameters	Value
Elitism	0.500
Mul.Probability	0.100
Mutation Rate	0.100
Crossover Rate	0.800
Population Size	50

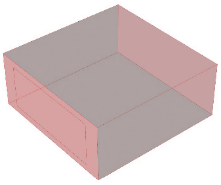
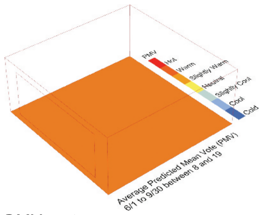
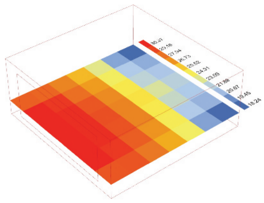
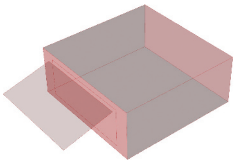
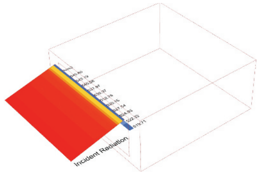
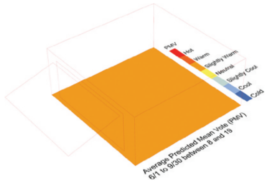
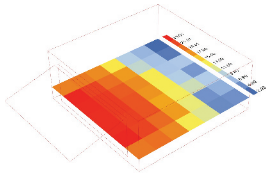
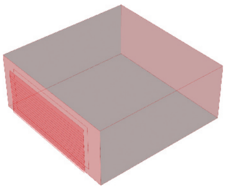
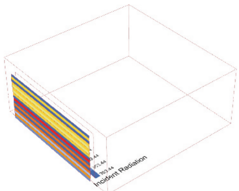
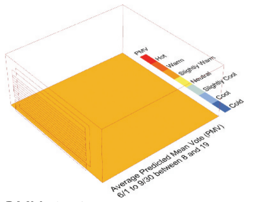
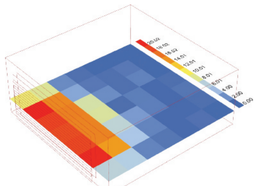
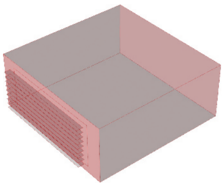
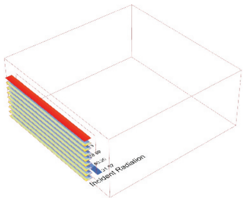
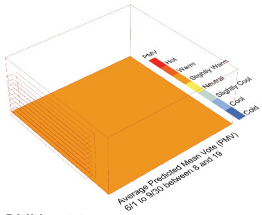
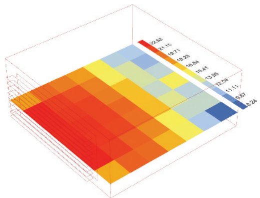
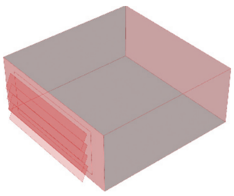
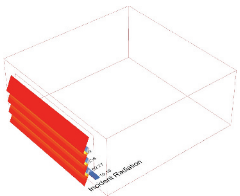
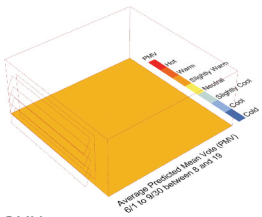
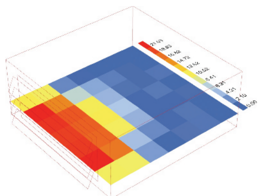
maximum evolution algebra. Table 6 shows the setting of input parameters in GA.

2.5. *Multi-criteria decision*

Unlike single-objective optimization, where the value of the optimal solution is the only one, multi-objective optimization is the selection of the optimal solution under the comprehensive consideration of multiple constraints. Therefore, various solutions can be obtained under different preferences, indicating that the optimal solution is not unique. The multi-criteria decision-making method is a decision-making method used to select the best plan among several plans that are conflicting and restricted by multiple attributes.

There are more than a dozen methods to optimize the solution set for multi-criteria optimization, including the Weighted Sum Method, Analytic Hierarchy Process (AHP), and Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) (analysis 2015). In decision theory, the TOPSIS method is the most used method for solving multi-objective decision-making. TOPSIS is a decision-making algorithm proposed by Hwang and Yoon (Tzeng and Huang 2011) in 1981, also known as the superiority and inferiority distance decision-making method, which is a ranking method that approximates the ideal solution by calculating the distance of each solution to the positive and negative ideal points and calculating the degree of proximity of each solution to the ideal solution, thus evaluating the relative superiority and inferiority of each solution and determining the relatively optimal solution. The optimal program is finally determined.

Table 7. Design strategies and parameters.

Program type	Optimized model	Solar radiation on the surface of PV louvers	Indoor thermal environment	Daylight autonomy
Base model	 Non-shading	— —	 PMV: 1.6 Discomfort Hours: 810h	 Cooling Load: 2879.73kWh Lighting Energy Consumption: 218.21kWh Total Energy Consumption: 3097.95kWh
Maximum PV power generation	 Shade Counts: 1 Angle: 15°	 PV Power Generation: 1576.78kWh Energy Substitution Rate: 66.8%	 PMV: 0.59 Discomfort Hours: 606h	 Cooling Load: 2022.43kWh Lighting Energy Consumption: 269.34kWh Total Energy Consumption: 2291.77kWh
Minimum cooling load	 Shade Counts: 8 Angle: 85°	 PV Power Generation: 1043.18kWh Energy Substitution Rate: 45.5%	 PMV: 0.58 Discomfort Hours: 364h	 Cooling Load: 1857.44kWh Lighting Energy Consumption: 435.92kWh Total Energy Consumption: 2293.36kWh
Minimum lighting energy consumption	 Shade Counts: 9 Angle: 0°	 PV Power Generation: 545.41kWh Energy Substitution Rate: 23.7%	 PMV: 1.02 Discomfort Hours: 666h	 Cooling Load: 2058.62kWh Lighting Energy Consumption: 238.02kWh Total Energy Consumption: 2296.64kWh
Minimum indoor thermal discomfort hours	 Shade Counts: 4 Angle: 65°	 PV Power Generation: 1044.53kWh Energy Substitution Rate: 46.2%	 PMV: 0.59 Discomfort Hours: 363h	 Cooling Load: 1867.65kWh Lighting Energy Consumption: 392.24kWh Total Energy Consumption: 2259.89kWh

3. Results & discussion

3.1. Preliminary analysis of simulation results

In order to clarify the extent to which the PV shading devices on the exterior windows of the west façade affect the internal performance of the room from June to September, the raw performance of the target building was simulated and analyzed in this paper. The results are shown in Table 7. The specific performance data of the base model of the study case are as follows: the cooling load is 2879.54 kWh, the lighting energy consumption is 217.97 kWh, the number of indoor thermal discomfort hours is 810 h, and the PMV is 1.6.

The optimized research parameter analysis results are shown in Figure 4, which is in line with the expected effect of the PV sunshade system. The distribution density of the lines in the figure shows that the Pareto optimal solution is mostly distributed in configuration with a small number of louvers. According to the line distribution of the performance indicators in Figure 4, there is an antagonistic relationship between the performance indicators, and the values of each performance indicator fluctuate within a certain range.

3.2. Analysis of single objective optimization scheme

When solving multi-objective optimization problems, multiple performance control indicators are changed simultaneously, and the influence of design variables on a single optimization variable cannot be obtained. This section analyzes the relationship between different design options and a single optimization objective, and the change rule of design variables when a single performance index is optimal, and the specific data for each design option are summarized in Table 6.

3.2.1. PV power generation is maximum

The simulation results of PV louvers with maximum power generation performance are shown in Table 7. When the number of blades is set between 1 and 4, and the blade inclination angle is between 0° and 70°, the power generation of PV louvers in the Pareto solutions is relatively large, and the data fluctuation range is 900 to 1600 kWh. When the number of blades is 8 to 15, the corresponding PV power generation fluctuates between 400 and 1200 kWh regardless of the blade inclination angle. After sorting the results of the Pareto solutions for the power generation of PV louvers, it is found that when the number of leaves in the PV louver shading system is 1, and the blade deflection angle of this strategy is 20°, the power generation of PV louvers reaches the maximum, reaching 1576.78 kWh. The cooling load is 2022.43 kWh, lighting energy consumption is 269.34 kWh, and PV power generation can offset 68.8% of the total energy consumption, but the number of thermal discomfort hours is only reduced by 94 hours. Moreover, after comparing the data between photovoltaic power generation and total building energy consumption across all scenarios, we found that on average, PV power generation accounts for approximately 48% of the total energy demand for the building; and the percentage of PVSD design strategies that comply with the values of the Chinese regional standard (China Academy of Building Research Ltd GB/T 51350–2019 2019) on energy-saving design for nearly zero energy buildings reaches 23%, which is of great significance in realizing the energy-

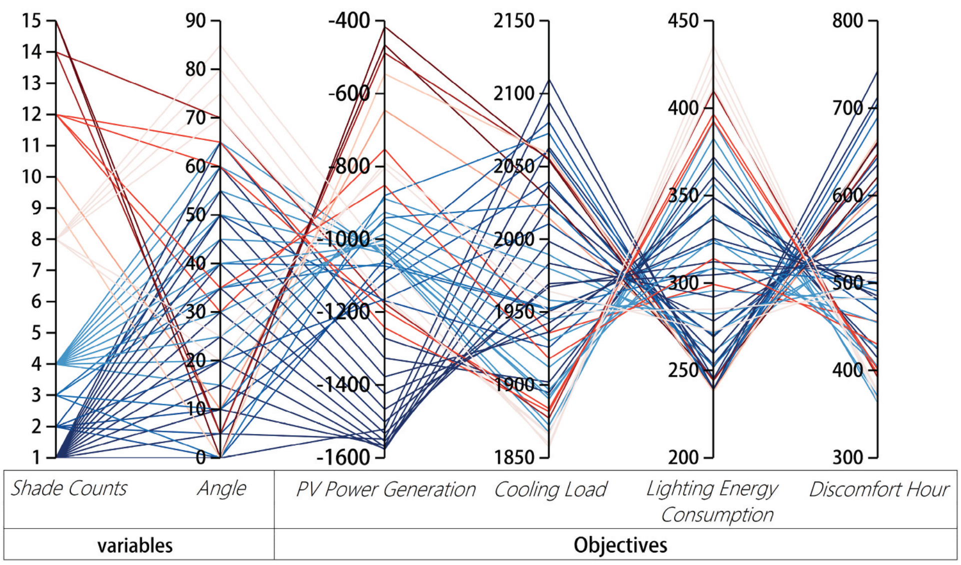


Figure 4. Parallel coordinate diagram of optimized design variables and optimization objectives.

saving and carbon-reducing efforts of the building industry.

3.2.2. Cooling load is minimum

According to the linear relationship between different design parameters and the cooling load in Figure 6, it can be found that the optimized building cooling load fluctuates between 1850 and 2110 kWh. According to the inclination of the lines and the degree of aggregation, it can be found that when the number of the blades is between 8 to 12 and the angle of the blades is between 30° and 70°, the building cooling load is smaller. When the number of blades is 1 to 3, the cooling load of the building increases with the increase of the deflection angle of the blades. After sorting the cooling load of Pareto solutions, it is found that when the number of blades is 8, and the inclination angle is 85°, the building cooling load is the smallest. This design scheme has a significant effect in reducing the number of thermal discomfort hours and cooling load, but the optimization potential in terms of PV power generation and lighting energy consumption is small.

3.2.3. Lighting energy consumption is minimum

Since the addition of shading devices will reduce the natural light level in the room and increase the degree of use of artificial lighting, it will increase the lighting energy consumption to a certain extent. Figure 6 shows that the change in lighting energy consumption is strongly related to the rotation angle of PV louvers, which increases with the increase of blade deflection angle. This is because the larger the blade angle is, the less natural light is obtained indoors, the worse the lighting quality is, and the greater the lighting energy consumption is. The number of blades is 9, the inclination angle is 0°, and the energy consumption of indoor lighting is minimal. Compared with the non-shading model, the lighting energy consumption of this solution increases by only 22 kWh, and the cooling load decreases significantly, with the PV generation offsetting 23.7% of the energy use, but the design strategy improves the indoor thermal environment to a lesser extent, reducing the discomfort hours by only 60 h, with an indoor thermal discomfort hourly rate of 666 h.

3.2.4. Indoor thermal discomfort hours are minimum

By observing the distribution of line colors in Figure 6, the results of indoor thermal discomfort hours under different design strategies are scattered, i.e., the changes of the two design variables significantly impact the indoor thermal environment. When the number of blades is 8, the greater the deflection angle of the blades, the better the improvement of the thermal environment; when the number of blades is 4, and the deflection angle of the louvers is between 15° and 65°, the improvement degree of the PV louver sunshade device to the interior is better; when the inclination angle is 65°, the number of indoor thermal discomfort hours is the lowest, only 363 hour.

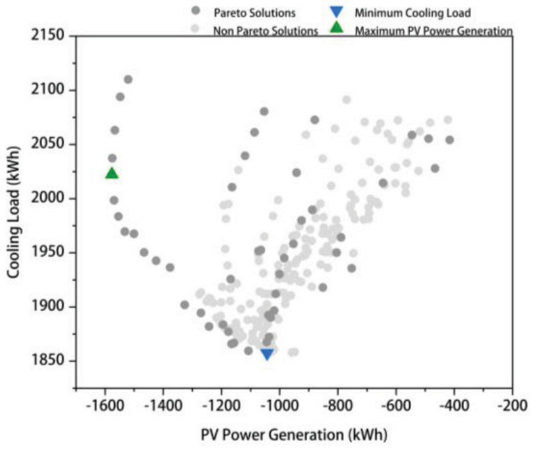
The discussion of Sections 3.1 and 3.2 reveals that as the number of PV louvers increases, i.e., the width of the blades increases, the PV power generation decreases. Therefore, we can conclude, similar to the results of previous studies (Sadatifar and Johlin 2022, Peres, Calili, and Louzada 2020), that the width of PV louvers is a key determinant of the energy efficiency of a building. In addition, the self-shading effect between PV louvers (Yadav et al. 2017) is one of the main reasons for the decrease in PV power generation with the increase in the number of blades, and it has been reported that the energy loss between PV panels due to the shading effect can reach up to 85%. Comparison of this study with the optimal energy saving solution for a single PV sunshade in Hong Kong[38], it is found that the energy saving rate of using PV louver is about 20% higher than that of single PV sunshade. Since office building cooling loads are the main cause of excessive summer energy consumption in buildings, the study was compared with the energy saving potential of the optimized design of the PVSD on the west façade (J. Liu, Bi, and ZHAO 2022) in Guangzhou, a city located in a region of China with hot summers and warm winters. It was found that the maximum PVSD power generation in both studies could offset more than half of the total building energy consumption.

3.3. Global analysis of optimization results

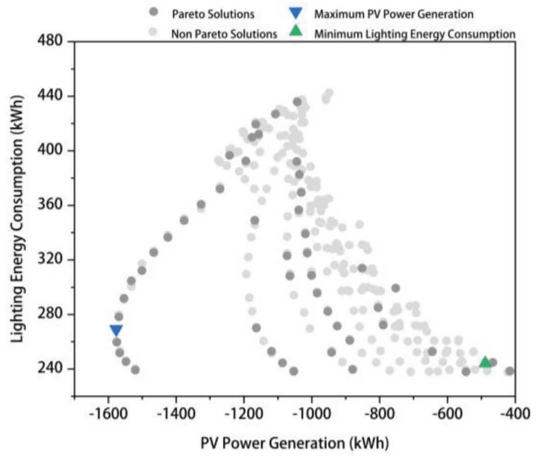
3.3.1. Analysis of the two objectives

In order to better understand the relationship between the objectives, the four objective function values of the optimized PV louvered shading system are transformed into a 2D (2-Dimensional) view by combining the two by two, as shown in Figure 5. The four objective function values are against each other. When one objective is optimal, the performance of other objectives may be the worst. Due to the conflicting checks and balances among the four objectives, their Pareto objective function values are not fully distributed on the boundaries of the region when mapped onto the 2D view.

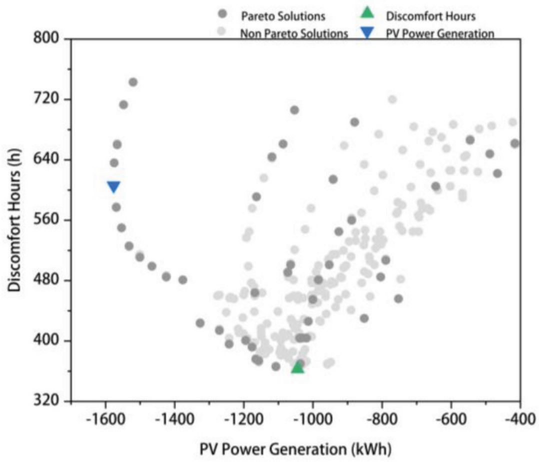
Figures 5(a)-(c) show the relationship between the power generation of PV louvers and the objective functions such as cooling load, lighting energy consumption, and indoor thermal discomfort hours for different optimization strategies. Figure 5(a) shows that with the reduction of building cooling load, the power generation value of PV louvers will stabilize in the range of 1000 – 1400 kWh, while Figure 5(b) shows that with the reduction of building lighting energy consumption, the power generation of PV louvers



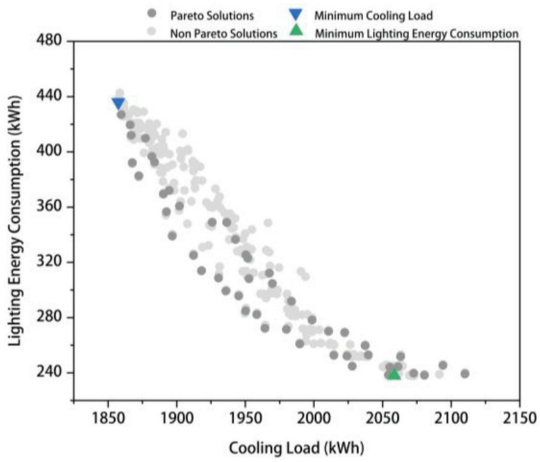
(a) PV Power Generation VS Cooling Load.



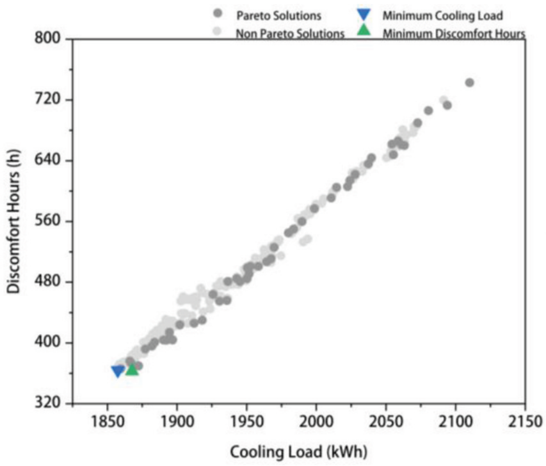
(b) PV Power Generation VS Lighting Energy Consumption.



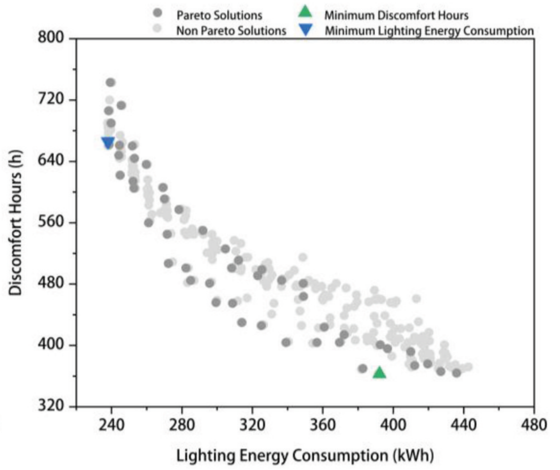
(c) PV power Generation VS Discomfort Hours.



(d) Cooling Load VS Lighting Energy Consumption.



(e) Cooling Load VS Discomfort Hours.



(f) Lighting Energy Consumption VS Discomfort Hours.

Figure 5. 2D mapping of the two optimization objectives.

fluctuates widely. As shown in Figure 5(c), when the number of indoor thermal discomfort hours is reduced to a certain level, the value of power generation of PV louvers also fluctuates between 1000 – 1400 kWh. Compared with the other two targets, the power generation of PV louvers and energy consumption of

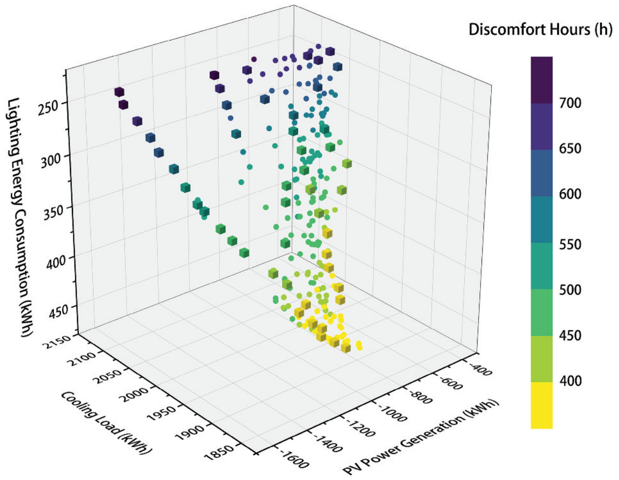


Figure 6. 4D mapping of the four optimization objectives.

building lighting in Figs. 5(a)-(c) has a larger range of numerical fluctuations, indicating that changes in the design variables of PV louvers have a greater impact on PV power generation and energy consumption of building lighting.

Figures 5(d)-(f) show a clear positive correlation between cooling load and indoor thermal discomfort hours, while lighting energy consumption negatively correlates with indoor thermal discomfort hours and building cooling load. In addition, the distribution of solution sets in Figure 5(e) and 5(f) reflects the relationship between cooling load and lighting energy consumption and between lighting energy consumption and thermal discomfort hours. Since Pareto solutions are always in the lead, coordination between conflicting objectives should be done when planning decisions are made.

3.3.2. Overall analysis of pareto solutions

Through the discussion in the previous section, the degree of influence between the four objective functions is clarified. This section discusses the interaction between the four objectives. The four-objective optimization solution set can be represented in a 3D space map to determine the results achieved concerning multiple objectives. The distribution of the solution set in space is shown in Figure 6, where the cube represents the Pareto frontier formed by the Pareto boundary solution, and the space is defined by PV Power Generation (X-Axis), Cooling Loads (Y-Axis), Lighting Energy Consumption (Z-Axis), and Discomfort Hours (Color-Axis).

Each objective function has its own magnitude. In order to unify the different magnitude and order of magnitude indexes in a unified dimension, this paper adopts the min-max Normalization method (Min-max Normalization) to carry out the reverse normalization processing. The index values of the four index parameters of the processed Pareto optimal solution are distributed in the interval [0–1], and each objective value is the closer to 1, the better. After normalization, the standardized indices of the objective function are plotted in the radar chart, as shown in Figure 7.

According to the distribution area in the four directions in Figure 7, it can be seen that most of the area is skewed towards the PV louver power generation and discomfort hours, indicating that both of them have a wide range of degrees of variation when optimization is carried out. Figure 7 shows that the shape of the area distribution of each group of optimization solutions in each direction fluctuates a lot, which likewise confirms the need to take into account the comprehensive

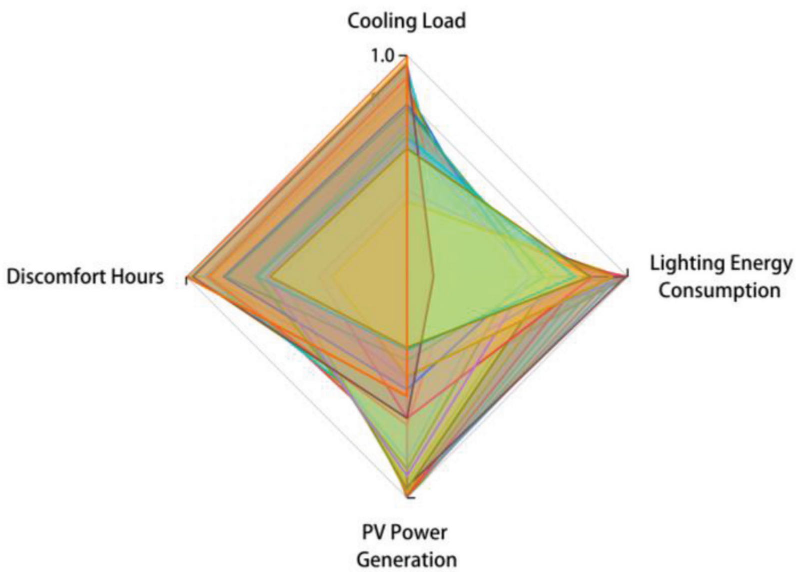
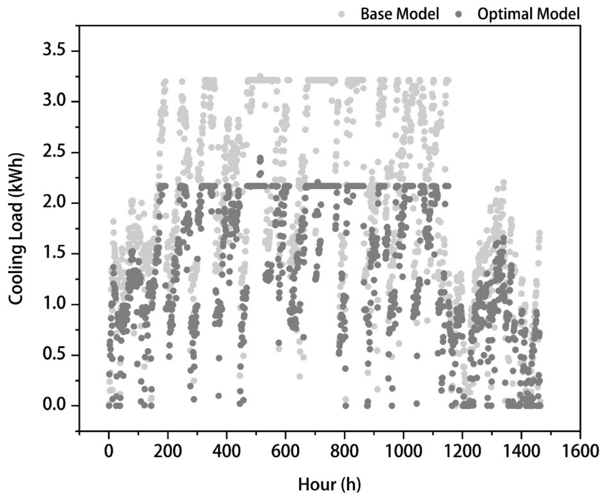
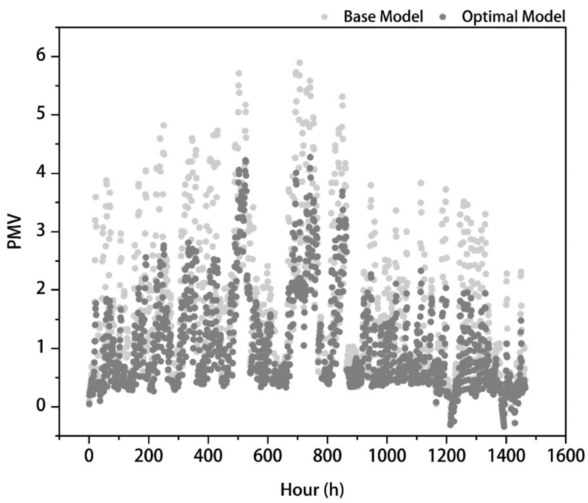


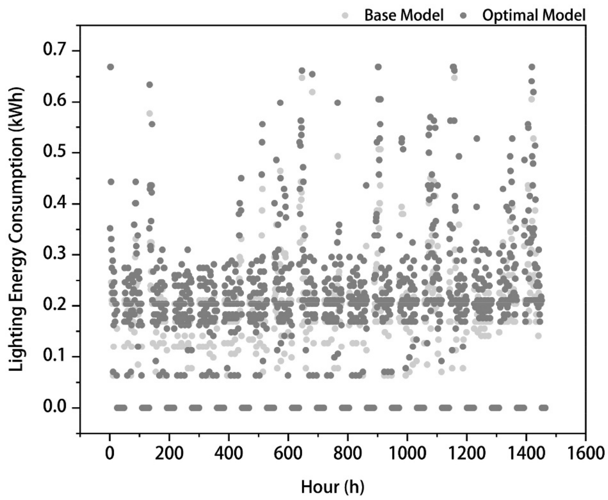
Figure 7. Radar chart after four-objectives normalization.



(a) Comparison of cooling load before and after optimization.



(b) Comparison of indoor thermal environment before and after optimization.



(c) Comparison of lighting energy consumption before and after optimization.

Figure 8. Comparison of room performance before and after optimization.

Table 8. Weighting coefficients for each objective.

PV power generation	Cooling load	Lighting energy consumption	Discomfort hours
0.22	0.26	0.26	0.26

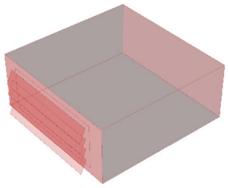
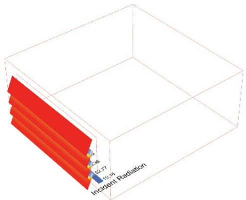
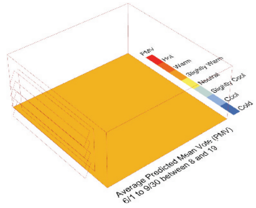
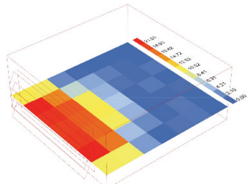
impact of each objective when performing multi-objective optimization.

It can be seen from the degree of intersection of the curves that when the line distribution between building cooling load and thermal discomfort hours tends to be parallel, the relationship between the two is positively correlated. In contrast, the intersection between thermal discomfort hours and PV power generation, cooling load and lighting energy consumption is relatively dense. The slope is large, indicating a relatively large negative correlation between the two targets.

3.4. *Pareto optimal solution selection*

This study discusses the distribution of PV louvers on the west façade when the four objectives of cooling load, lighting energy consumption, thermal discomfort hours, and PV louver power generation are optimized. Regarding the selection of design parameter combinations for PV louvers, decision makers can set the weight of performance indicators according to different needs and then obtain the design strategy of PV louvers shading that meets the design tendency. The Entropy Method is an objective weighting method, and its biggest feature is to directly use the decision matrix to calculate the weight, making the evaluation result more scientific (Huo 2021).

Table 9. Optimal program parameters.

Program type	Optimized model	Solar radiation on the surface of PV louvers	Indoor thermal environment	Daylight autonomy
Optimal Strategy	 Shade Counts: 4 Angle: 45°	 PV Power Generation: 1019.19kWh Energy Substitution Rate: 44.44%	 PMV: 0.63 Discomfort Hours: 404h	 Cooling Load: 1857.44kWh Lighting Energy Consumption: 435.92kWh Total Energy Consumption: 2293.36kWh

In this paper, the entropy weight-superior distance (TOPSIS) method was used to comprehensively evaluate and sort the non-dominated solutions after 30 sets of Pareto normalization processing obtained in this optimization design to select the best design scheme. After using the entropy weight method to repeatedly assign weights to the four objectives of this study, the weight results are shown in Table 8:

After obtaining the weights of each objective by the entropy weighting method, the Pareto frontier solution was ranked using the TOPSIS multi-criteria decision-making method. The TOPSIS multi-criteria decision-making results are shown in Table 8. After multi-criteria decision-making on the Pareto solutions, the number of PV louver blades in the optimal design is 4, the blade tilt angle is 45°, the number of uncomfortable hours in the hot room is 404 h, the total energy consumption of the building is 2293.36 kWh, and the PV power generation is 1019.19 kWh, which can offset 44.44% of building energy consumption. The specific data are shown in Table 9.

A comparison of the optimized scheme with the performance of the base model building is shown in Figure 8, and the scatter distribution in Figures 8(a) and 8(b) reveal that the optimized strategy can significantly reduce the indoor high-temperature situation and the peak cooling load of the building from June to September in summer to maintain good comfort indoors. However, the comparison graph of the indoor lighting energy consumption in Figure 8(c) reveals a slight increase in the lighting energy consumption of the building after the optimization.

4. Conclusion

This paper proposes a simple, practical, and efficient multi-objective optimization design platform for installing PV louvers on the west facade of a building. Taking an office building in Zhengzhou, China, as an example, the optimal configuration of the external solar shading device for west-facing PV louvers is optimized and designed, aiming to improve the utilization of solar renewable energy while reducing building energy consumption and the phenomenon of thermal discomfort. The optimization framework uses Rhino software to model the building, the Ladybug & Honeybee plug-in in Grasshopper to simulate the energy and lighting environment of the office with windows set in the west façade, and the plug-in Octopus for optimization. To further determine the optimal system configuration that meets the objectives, the TOPSIS method is introduced to perform multi-criteria decision-making on the Pareto solutions to obtain the optimal design strategy that meets the decision maker's preference and to provide the designers with references and bases for design choices.

In the Pareto solutions, the design strategy with a small number of PV louver blades accounts for a larger proportion than the design strategy with many blades. In the distribution of the number of louvers, the distribution of the number of leaves in the optimal solution is clear and dense. The variation range of the blade deflection angle is comprehensive, and its transformation degree covers 0° to 85°. Hence, the inclination angle of the PV louver has a greater impact on various performance indicators.

In selecting the optimal solution, the article adopts the entropy method to objectively weigh each target weight and uses the TOPSIS method to screen the optimal solution. Compared with the baseline model, the lighting energy consumption has slightly increased, but the cooling load has dropped by nearly 1000 kWh, and the total energy consumption has dropped by about 800 kWh. In addition, designers can give different weights to the optimization objectives according to their preferences to get more optimization solutions, which is significant for the early design stage of buildings.

Future research could be further optimized to eliminate the limitations of this current study. Firstly, in order to simplify and restrict the research content, the paper aims at a single room in the simulation of the study case, without considering the impact of the surrounding building environment on building performance. Secondly, the study only analyzed the degree of influence of design variables on building performance under a single climatic condition, and it is necessary to expand the study to incorporate different seasons and different typical climatic zones in the future in order to enhance the broad applicability of the research methodology. Moreover, the incorporation of light environment quality evaluation criteria into the optimization objectives should be explored to mitigate the visual impact of solar radiation on indoor occupants.

Disclosure statement

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